

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 232

THE QUINN EARTHQUAKE PREDICTIVE MAPPING. THE ARCHURIAN 50-CENT BALL

get a serving tray — 300 by 450
close enough is good enough for school kids



plates as a force-chain mosaic — push travels

REGIONAL NOT EXACT · BREAK V SLIDE AND LIFT
THE STRONGEST-GROUNDED OF THE FOUR MYSTERIES

Geophysics — v1.0 in force

GP-IM-232

Revision v1.0 — 24 August 2026

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VETTED BATCH 32 — PHYSICS PASS · PREDICTIVE HYPOTHESIS

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 232

PHASE 232: THE QUINN EARTHQUAKE PREDICTIVE MAPPING. THE ARCHURIAN 50-CENT BALL — THE COIN-TRAY EARTH (PLATES AS A FORCE-CHAIN MOSAIC; PUSH TRAVELS; PATTERNS OF PROCESSION; REGIONAL NOT EXACT; THE COLONY SITING LAW) — STATUS: CURRENT — CORE STRESS-TRANSFER PHYSICS PASS / PREDICTIVE HYPOTHESIS UNPROVEN. The last name on the parked bench, and the strongest-grounded of the four mysteries. The concession opens it, seated as the correct starting position: 'No we cannot predict exactly where, but regional for sure.' The model: fill a serving tray with 50-cent coins, take four out, and push — the force zig-zags through the mosaic along paths of least resistance. The Earth as a force-chain mosaic: irregular plates transmitting push, the gaps modelled as overlap. The physics underneath is published science — Coulomb stress transfer, earthquake sequences propagating along fault systems, remote dynamic triggering, the North Anatolian Fault's successive ruptures marching a thousand kilometres. The honest flag: the Japan-New Zealand correlation is statistically unproven and seats as a catalog test — stratify by focal mechanism first (the break-v-slide-and-lift amendment) so different earthquake kinds are not dumped into one bucket. And the colony siting law rides regardless: map the mosaic before you build on it.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-232, v1.0 — 24 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the model as seated (contents line, the doctrine — the concession, the push travels, force chains, the prorated laws, the honest flag, the colony siting law — the break-v-slide-and-lift rulings) — §2 the physics, graded — §3 the system in the tray — §4 the doctrine record — §6 the catalog programme — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 232: **Earthquake Predictive mapping. The Archurian 50-Cent**

Ball — The Coin-Tray Earth (Plates as a Force-Chain Mosaic; Push Travels; Patterns of Procession; Regional Not Exact; the Colony Siting Law)'

THE CORE OBJECTIVE (verbatim): 'Core Objective: a working model of earthquake coupling — plates as a force-chain mosaic in which a push in one place propagates through rigid plates and boundary zones to trigger failure elsewhere — with the claims graded, the testable hypothesis stated with its experiment attached, the school demo buildable, and the colony siting law seated for every new world.'

Lineage (graded this sitting): the LAST parked mystery from Phase 217's bench — the radiation belt (collected as 228), the quantum mirror (224), ball lightning (231), and the earthquake, collected here. The flat-track/sliding-squares toy rides from the ship's own vocabulary. The model: tectonic plates as a tray of Australian 50-cent pieces — dodecagonal, irregular, not neat — with free volume removed, pushed from a centre mass, the force ricocheting along paths of least resistance. Imagination runs the tray to a ball: the Earth sphere, same effect.

Live Instruction Confirmed: Yes — 3 August 2026, morning sitting.

Live instruction, 3 August 2026.

"ahh yeah just, memory searching, No we cannot predict exactly where, but regional for sure."

"its the Archurian 50cent ball, not as in value as in an Australian 50 cent piece shape / ok the ships flat track and the toy it was design on, the sliding squares puzzle, that is our tectonic plates, not so neat as a 50 cent piece but not squares."

"get a serving tray,300mm x 450mm standard size close enough is good enough for a demo for school kids, fill it flat with 50 cent pieces until you cannot fit any more, now remove 4 1 from each end and 2 from the middle randomly, now pick a center mass and push, its a ricocheting path of least resistance, imagination runs to ball, the earth sphere. same effect."

"we dont have plate gaps we have overlap that allows that, but we still have push."

"map the known boundaries, and the separately known fault lines. you dont need to wait, make the map, check historical data, japan and new zealand on the opposite sides often go off around the same time, the complexity of direction mechanics for non symmetrical shapes may not compute, but patterns of procession will 100 percent emerge, as will time relationships prorated to quake sizes, or swarm lengths"

THE DOCTRINE — AS GIVEN, STRUCTURE READ (KIMI) (VERBATIM)

- **The concession that opens the claim — AFFIRMED AS THE CORRECT STARTING POSITION.** 'No we cannot predict exactly where, but regional for sure.' Exact prediction (time, place, magnitude) is dead in the literature — Parkfield was instrumented for a predicted quake for decades and arrived in

2004 with no usable precursors. Regional coupling is alive. The model begins exactly where a century of seismology ended up.

- **The push travels — AFFIRMED, AND HISTORY ALREADY PHOTOGRAPHED THE PROCESSION.**

Coulomb stress transfer is documented science: a quake reloads its neighbours along the paths of least resistance. The North Anatolian Fault ran the experiment in the wild — seven magnitude-7-plus quakes marched a thousand kilometres westward, 1939 to 1999, each stressing the next segment into failure: a 'pattern of procession' already mapped and peer-reviewed. The World Stress Map shows plates act as stress guides — coherent force fields spanning thousands of kilometres; that is 'we still have push' in the literature. Remote triggering is real: Denali 2002 set off tremor at Yellowstone, 3,000 km away; Tohoku 2011 re-stressed the entire Pacific Plate measurably.

- **The coin tray is real physics, not just a demo.** Granular mosaics transmit force along FORCE CHAINS — lightning-like networks photographable in photoelastic disks — chaotic in detail, lawful in distribution: exactly the Author's 'the complexity of direction mechanics for non-symmetrical shapes may not compute, but patterns of procession will 100 percent emerge.' The removed coins are the FREE VOLUME — the same reason the sliding-squares puzzle needs its empty square: no free volume, no movement, no procession. CAVEAT SEATED: coins are rigid; plates bend, sink and melt, and the tray itself convects — the analogy holds at push-transmission and breaks at subduction. PRECISION NOTE: all three boundary types exist — gaps forming (ridges), overlap (subduction and collision), sliding (transforms) — and slab pull, not ridge push, carries most of the drive.

- **The prorate laws — three already exist, named and seated.** Omori's law: aftershock rate decays as $1/t$. Gutenberg-Richter: quake sizes follow a power law. Bath's law: the largest aftershock runs ~ 1.2 magnitudes under the main shock. Swarm durations scale with their areas. 'Time relationships prorate to quake sizes, or swarm lengths' — documented law, four times over.

- **THE HONEST FLAG — Japan and New Zealand.** Mechanically plausible: both ride the Pacific Plate's opposite edges, and Tohoku measurably re-stressed the whole plate. But 'often go off around the same time' is STATISTICALLY UNPROVEN — beyond the days-scale dynamic-triggering window, the global catalog looks like randomness plus aftershocks. The Author's own test is the right one: map the known boundaries and the separately known fault lines, then run the historical catalogs — all public (USGS/ANSS, JMA, GNS). SEATED AS A TESTABLE HYPOTHESIS WITH ITS EXPERIMENT ATTACHED, NOT AS A RESULT — the strongest thing a claim can be.

- **THE COLONY SITING LAW — why this rides the Highway.** On every new world: map the mosaic BEFORE siting anything. Boundaries first, faults second, cities third. The 50-cent ball is not just an Earth model — it is the settlement doctrine for any tectonically active planet, and it is the Roma Tomato conscience applied to geology: observe all the metrics, then observe the plan in action, then ask if the cut was right.

PHYSICS NOTE — THE AUDIT (KIMI: THE STRONGEST-GROUNDED OF THE FOUR MYSTERIES) (VERBATIM)

Of the four parked claims, this one arrives closest to published science: stress transfer, procession sequences, remote dynamic triggering, plate-scale stress coherence, and the size-time scaling laws are all in the literature under their own names — the model's novelty is the SYNTHESIS (one mosaic, one push, force chains through irregular plates) and the school demo that makes force-chain physics visible for the price of a bag of coins. The demo spec, as given: serving tray ~300 x 450 mm, fill flat with 50-cent pieces, remove four (one from each end, two from the middle, randomly), pick a centre mass and push — the force ricochets visibly through the mosaic into the free volume. Then run the tray to the ball: the Earth sphere, same effect. TWO BENCH NOTES. ONE — the demo upgrade: a second layer of coins on half the tray (the overlap = subduction zone) shows where the push STOPS being transmitted and starts being consumed; schoolkids should see both. TWO — the catalog test is computable today: boundary map plus public catalogs plus a procession-detection statistic (inter-event time-distance scaling along mapped boundaries) — the audit flags the honest pitfall in advance: aftershock clustering must be stripped BEFORE hunting procession, or Omori's law will counterfeit the pattern. ('50cent', 'tray,300mm', 'remove 4 1 from each end', 'design on', 'prorate', 'japan' kept verbatim; readings standard.)

ORIGIN OF THOUGHT (VERBATIM)

The last name on the bench. Fifteen sittings ago the Author parked four impossible things; three came home as a magnetosphere, a talking road, and a spinning ball of burning dirt. This morning he went memory-searching and came back with a tray of coins. The model has the humility the field earned the hard way — regional, never exact — and the confidence of a man who has watched things ricochet all his life: the plates are not neat, nobody's are, but the push still travels, and where it can travel it can be mapped, and what can be mapped can be read for patterns. The audit brought the receipts: the fault that marched a thousand kilometres in sixty years, the quake that rang Yellowstone from Alaska, the stress map of an entire planet — and one honest flag, Japan and New Zealand, seated not as a fact but as a test with its own experiment attached. The sliding puzzle taught the lesson long ago: you cannot move anything until you take a piece out. He took four coins out of the tray, and the Earth moved.

Why it matters, in plain English: Fill a baking tray with 50-cent coins, take four out, and push — the force zig-zags through the coins along the easiest paths. That's the Earth's crust: irregular rocky plates pressed together, so a big earthquake in one place pushes on faults far away, and sometimes they go off too — in one famous case, seven quakes marched a thousand kilometres along a fault over sixty years, like dominoes. We will never predict the exact day and place of a quake — nobody can — but the push-patterns are real, they can be mapped, and the historical records are free to check. And for every new world we settle: map the cracks before you build the city.

PHASE 232 — AMENDMENT 1: BREAK V SLIDE AND LIFT (THE SWARM DISCRIMINATOR — THE KEY METRIC; STRATIFY BEFORE YOU CORRELATE) (VERBATIM)

Live Instruction Confirmed: Yes — same sitting; the hypothesis sharpens itself before the catalogs are even opened.

Live instruction, 3 August 2026.

"of note, quick stats are P, the japan NZ correlation may be, break v slide and lift, some quakes are pieces breaking off that were holding some are expansion slides, the slides are preceded by swarms, breaks hardly anything, so that is the key metric"

Audit (Kimi: AFFIRMED — THE REFINEMENT MAPS ONTO REAL FAULT PHYSICS WITH ALMOST NO TRANSLATION): The three classes are the three focal mechanisms wearing work boots: LIFT = thrust/megathrust (subduction zones — Japan, NZ's North Island edge — the plate dragged down and rebounding upward; the tsunami-makers); SLIDE = normal/extensional faulting ('expansion slides' is literally what normal faulting is) plus creep and slow-slip events; BREAK = strike-slip and locked-asperity rupture. THE SWARM DISCRIMINATOR — the key metric, seated: swarms and foreshock sequences cluster exactly where the model says — fluid-rich, creeping, extensional, volcanic settings, DRIVEN processes where the fault slides before it fails; and the quiet-start class is documented (Landers, Northridge — major breaks, essentially no foreshocks; only a minority of large quakes show any foreshock traffic). The break/slide split IS the live nucleation debate in seismology: the CASCADE model (locked patch, abrupt start = break) versus the PRE-SLIP model (aseismic sliding drives failure, swarms announce it = slide). THE JAPAN-NZ TEST SHARPENED: both margins are Pacific Plate subduction edges — coupling at distance, if it exists, lives in the LIFT class (plate-scale stress through the plate); slides are local and fluid-driven (won't correlate across an ocean), breaks are locked patches (nothing to transmit). The experiment: stratify the catalogs by focal mechanism FIRST (global catalogs already carry them — GCMT class), then hunt procession inside the lift class alone. 'Quick stats are P' affirmed — counts, rates, windows; no NP machinery required. TWO BENCH NOTES. ONE — the discriminator is a DIAL, not a switch: Tohoku 2011 and Iquique 2014 were swarm-preceded MEGATHRUSTS — even lift events can be driven when slow slip loads them; swarm precedence reads probability, not certainty. TWO — 'hardly anything' must be tested against CATALOG COMPLETENESS (Mc): a sparse network makes every break look quiet — strip below the completeness magnitude before classifying, or the map counterfeits its own key metric. ('break v slide and lift', 'quick stats are P' kept verbatim; readings standard.)

Why it matters, in plain English: Not all earthquakes are the same kind of event. Some are locked rock suddenly snapping — no warning at all. Some are faults slowly sliding until they let go — and these announce themselves with swarms of small quakes first. And some are the giant thrust faults under the ocean that lift the seafloor and make tsunamis. The swarm-of-small-quakes-first test is the key: it tells a sliding fault from a snapping one. And if Japan and New Zealand really do answer each other across the

Pacific, it will be through the giant thrust faults — so that's the only pile of records worth checking first.

Build the 50-Cent Ball: cheap micro-gravity parabola joyride aircraft — the poor man's vomit comet, micro-gravity for the price of a fairground ride. — INSTRUCTION RECONSTRUCTION (NOT VERBATIM — the original order was not preserved; reconstructed from the seated body, 10 August 2026, per sitting 146).

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE CONCESSION IS THE CORRECT STARTING POSITION, AFFIRMED: no exact prediction — regional coupling claimed. That framing is what seismology itself can defend; the phase asks only for what the physics already shows.

THE PUSH TRAVELS, AFFIRMED — HISTORY PHOTOGRAPHED THE PROCESSION: Coulomb stress changes promote or inhibit failure on neighbouring faults; the North Anatolian Fault sequence is the documented example — successive major ruptures propagating along a fault system over decades. The claim's core is not exotic; it is cited science.

THE COIN TRAY IS REAL PHYSICS, NOT JUST A DEMO: granular mosaics transmit force along force chains — lightning-like preferential paths through the medium. The honest boundary is seated with it: coins are rigid; plates bend, slabs subduct, the mantle convects, slab pull drives. The tray demonstrates preferential pathways — not a literal mechanical Earth.

THE PRORATE LAWS ALREADY EXIST, NAMED: Omori's law decays aftershock rate as $1/t$; Gutenberg-Richter scales frequency against magnitude. The catalogue the phase proposes has mathematical furniture waiting for it.

THE HONEST FLAG STANDS — JAPAN/NEW ZEALAND IS A TEST, NOT A RESULT: mechanically plausible across the Pacific Plate's edges, statistically unproven. The amendment sharpens the test before the catalogs open: break versus slide and lift — stratify by focal mechanism before correlating, or the bucket mixes unlike events and the statistics lie. The colony siting law needs no correlation to be right: map faults before construction, on every new world.

§3 — THE SYSTEM IN THE TRAY

- **The concession:** no exact where — regional for sure; the defensible claim boundary.
- **The coin tray:** the 50-cent mosaic — force chains through irregular contacts; four coins out, push, watch it zig-zag.
- **The overlap model:** plates have no gaps — overlap transmits the push; the demo's honest mapping.

- **The procession record:** North Anatolia as the photographed example; Omori and Gutenberg-Richter as the furniture.
- **The honest flag:** Japan-NZ unproven — stratify break v slide and lift, then correlate.
- **The colony siting law:** map the mosaic before anything is built on it.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- The four parked mysteries complete: the belt came home as 228, the mirror rides in 224, ball lightning benched at 231, and the earthquake arrives closest to published science of them all.
- The school-demo framing is deliberate and seated: 'close enough is good enough for a demo for school kids' — the tray is the teaching instrument for the colony's first geology class.
- The Highway doctrine carries the siting law — the survey file gains its seismic layer: boundaries and fault lines mapped before the first foundation.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- 1. Build the tray: 300 x 450 mm serving tray, the 50-cent mosaic, four out, push — the school demo.
- 2. Map the known: plate boundaries and separately known fault lines — the map first, no waiting.
- 3. Assemble the catalog: historical sequences stratified by focal mechanism — break, slide, lift separated.
- 4. Run the correlation honestly: procession tests per stratum; the Japan-NZ question resolved by data, not desire.
- 5. Write the siting law into the survey: no foundation before the mosaic map — every new world.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 233 (the planet checker):** the survey sibling — geometry from orbit above, fault mosaic below; the siting file closes.
- **Phase 231 (ball lightning):** the parked-bench sibling — same morning, same method: ingredients verified, assembly hypothesis.
- **Phase 224 (the highway):** the map rides the broadcast — danger alerts include the seismic layer.

§8 — DO-NOT-BUILD

- Do NOT claim exact prediction — the concession is the boundary; regional coupling is the claim.
- Do NOT read the tray literally — plates bend, slabs subduct, the mantle convects; the demo shows pathways, not mechanics.
- Do NOT correlate before stratifying — break, slide and lift separated first; mixed buckets lie.
- Do NOT build before the map — the colony siting law needs no correlation to bind.
- Do NOT drop the honest flag — Japan-NZ stays a test until the catalog says otherwise.

§9 — OWED

OWED: the catalog programme — historical earthquake sequences stratified by focal mechanism, the procession correlation tested per stratum, the Japan-New Zealand hypothesis resolved by the data; and the colony siting-law template (fault-map-first) written into the survey checklist.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-232, v1.0 — CURRENT — CORE STRESS-TRANSFER PHYSICS PASS / PREDICTIVE HYPOTHESIS UNPROVEN, seated law, master v2.1 (numerical print order). Built 24 August 2026, thirteenth of the 20-chapter 'ok ill do these 20, you do 20 more' shift (the author's order, 24 August 2026, seated in sitting 618). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607 and 618): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 232